

USE CASE-BASED LEARNING TO RECOMMEND TREATMENT PLANS BASED ON SIMILAR PATIENT HISTORIES

Relevant SDG: SDG 3 – Good Health and Well-Being

1. PROBLEM STATEMENT

Healthcare professionals often need to select suitable treatment plans based on a patient's symptoms, medical history, age, laboratory results, and previous treatments. Manually comparing a patient's history with similar previous cases can be time-consuming.

This project proposes a **Case-Based Learning (CBL)** system that identifies patients with similar medical histories and recommends a suitable treatment plan based on the outcomes of those previous cases.

2. SDG RELEVANCE

SDG 3 – Good Health and Well-Being

The project supports SDG 3 by using machine learning to assist healthcare decision-making, improve treatment recommendations, and potentially support better healthcare outcomes.

3. DATASET DESCRIPTION

A suitable healthcare dataset can be obtained from **Kaggle or the UCI Machine Learning Repository**.

Possible attributes:

Feature	Description
Age	Patient age
Gender	Patient gender

Feature	Description
Blood Pressure	Patient blood pressure
Blood Glucose	Glucose level
BMI	Body Mass Index
Symptoms	Patient symptoms
Diagnosis	Diagnosed condition
Previous Treatment	Earlier treatment
Treatment Outcome	Outcome of treatment

Target: Recommended treatment / treatment outcome.

4. DATA PREPROCESSING

The dataset will be preprocessed using the following steps:

- Remove duplicate records.
- Handle missing values.
- Convert categorical variables into numerical values.
- Normalize numerical features.
- Remove irrelevant attributes.
- Divide the dataset into training and testing data.

5. MODEL DEVELOPMENT – CASE-BASED LEARNING

Case-Based Learning works by comparing a new patient's information with previously stored patient cases.

Process:

1. Store historical patient cases.
2. Receive the new patient's information.
3. Calculate similarity between the new patient and previous cases.

4. Identify the most similar cases.
5. Analyze the treatments and outcomes of those cases.
6. Recommend the treatment associated with the most successful similar cases

A similarity measure such as **Euclidean distance** can be used for numerical features.

6. MODEL EVALUATION

The model can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

If the system predicts a numerical treatment outcome, **MAE and RMSE** can also be used.

7. RESULT VISUALIZATION

The following visualizations can be included:

- Patient age distribution
- Diagnosis distribution
- Treatment distribution
- Confusion matrix
- Accuracy/Precision/Recall/F1-score comparison
- Actual vs. predicted treatment outcomes
- Similarity distribution of retrieved cases

8. INTERPRETATION OF RESULTS

The Case-Based Learning model identifies historical patients who have similar characteristics to a new patient. If the retrieved cases show successful outcomes for a particular treatment, that treatment can receive a higher recommendation score.

For example:

If a new patient has characteristics similar to several previous patients who responded successfully to Treatment A, the system will recommend Treatment A as a potentially suitable option.

Important: The system should be presented as a **decision-support tool**, not as a replacement for a qualified healthcare professional.

9. SDG-BASED RECOMMENDATIONS

Based on the results, the following recommendations can be made:

1. Use historical patient data to support faster clinical decision-making.
2. Use similarity-based models to identify patterns among successful treatment cases.
3. Continuously update the case database with new patient outcomes.
4. Ensure patient privacy and secure handling of medical data.
5. Use the model as decision support alongside professional medical judgment.

10. CONCLUSION

This project demonstrates how **Case-Based Learning** can be applied to healthcare data to identify similar patient histories and support treatment recommendations. By reusing knowledge from previous cases, the system can provide useful insights for healthcare decision-making. The project contributes to **SDG 3 – Good Health and Well-Being** by exploring machine-learning techniques that can support improved and efficient healthcare services.

11. REFERENCES

1. UCI Machine Learning Repository – Healthcare datasets.
2. Kaggle – Healthcare datasets.

3. World Health Organization (WHO) – Health-related datasets and resources.
4. United Nations – Sustainable Development Goal 3: Good Health and Well-Being.
5. Research literature on Case-Based Reasoning and Instance-Based Learning.